

BIPUL CHANDRA DEV NATH

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Professional Summary

Senior Full-Stack Developer and Researcher with 15+ years of experience architecting, developing, and deploying complex, high-impact applications. Deep, hands-on expertise in building scalable **e-learning platforms** and **AI-driven MedTech solutions**. Proven ability to lead the full project lifecycle, from initial R&D and concept to deployment and long-term maintenance.

- **Specializations:** PHP (Laravel, CodeIgniter), Python (AI/ML), Flutter, and Java.
- **Core Competencies:** Full-Stack Development, Academic & Applied Research, AI/ML Integration (scikit-learn, TensorFlow), Database Architecture (MySQL, MongoDB, PostgreSQL), and Cross-Platform Mobile Development.
- **Key Achievements:**
 - Developed and currently maintains an SQL learning platform (dbles.esrdlab.org) actively used by North South University (NSU) faculty and students for **over 4 years**.
 - Architected a novel AI-powered web application for the early-stage detection of CVD, diabetes, and nephrology diseases.
 - Led the development of a pioneering online e-learning platform (hscict.org) for HSC ICT students, featuring live compilers for C, HTML, and SQL, eliminating the need for local installations.

Technical Skills

Category	Technologies
Expert Level	PHP, Laravel , Codeigniter , MySQL, Git
Proficient: Full-Stack	Python (Django), JavaScript (ES6+), React (MERN), Node.js, Java, HTML5, CSS3/SCSS
Proficient: Mobile	Flutter , Dart, Objective-C, Unity3D (C#), Cocos2D
AI & Machine Learning	scikit-learn , TensorFlow , Natural Language Processing (NLP), Pandas, NumPy
Databases	MySQL , MongoDB , PostgreSQL , SQLite, SQFlite , Supabase
Concepts & Tools	R&D, System Architecture, REST APIs, JSON, UML, Agile Methodologies

Work Links

<https://bdmeditech.com/>
<https://hscict.org/>
<https://ugradc.3dotsss.com/>
<https://dbles.3dotsss.com/>
<https://bdjobsguru.com/>
<https://www.timepiecebd.com/>
<https://barberschoolbd.com/>
<https://vinternship.org/>
<https://gita.bdjobsguru.com/>
<https://play.google.com/store/apps/developer?id=Bipul+Chandra+Dev+Nath>

Full-stack developer responsible for end-to-end development and ongoing maintenance of the above web applications.

Professional Experience

Senior Developer & Researcher | Institutes of Information and Communication Technology (IICT), BUET

Dhaka, Bangladesh | 2015 – Present

Senior developer and core researcher for a university-affiliated lab (esrdlab.org), focused on architecting and deploying grant-funded, high-availability e-learning platforms and novel health AI systems.

- **AI-Powered Medical Diagnostic Tools (2025)**

- Chronic Disease Prediction & Patient Management**

- Architected and developed a web application to predict chronic disease probability (CVD, diabetes, nephrology) using AI models (scikit-learn).
 - Developed a companion cross-platform Flutter application to help patients manage their health information, linking directly to the AI diagnostic tool.

- Radiological AI Detection**

- Built and deployed an AI tool using Flask to detect and classify knee arthritis from patient X-ray images.

- **SQL Learning and Evaluation System (SQL-LES) (2017-Present)**

- Led the development of the second-generation dbles.esrdlab.org, migrating the platform from Codeigniter to **Laravel** for improved performance and scalability.
 - The platform (v1 & v2) has been in continuous use by North South University (NSU) faculty and students for **over 4 years**, facilitating database query practice, automated exam evaluation, and schema-based learning.
 - *Role: Full-stack development, database architecture (prebuilt schemas), and system maintenance.*

- **Blockchain-Based Patient Record Linkage (BlockPRLS) (2022)**

- Served as lead developer for a grant-funded project to create a secure, blockchain-based

system for patient record linkage, enabling big data analytics.

- Co-authored and published the resulting research in the 2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC).
- *Technologies: PHP, Flutter, and blockchain concepts.*

- **Problem-Based E-Learning (EPBL) for HSC ICT (2015-2017)**

- Architected and deployed hscict.org, a first-of-its-kind online e-learning platform for Higher Secondary (HSC) students in Bangladesh.
- Engineered the core feature: a web-based execution environment allowing students to run **C code, HTML, and database queries** directly in the browser, removing all barriers to installation.
- Successfully extended the platform for university-level C-programming courses.
- *Technologies: Codeigniter, PHP, MySQL, and custom-integrated compilers.*

- **Virtual Internship System (VIS) (2020-2021)**

- Contributed to the conceptual design and development of a virtual internship system to benchmark software development skills in a blended learning environment.
- Co-authored and published the concept in the 2020 23rd International Conference on Computer, Information Technology (ICCIT).

Lead Developer |

Dhaka, Bangladesh | c. 2013 – 2015

- Developed a startup and successfully acquired clients for custom-built solutions.
- Designed, developed, and deployed a complete **Java-based product management application** for medical stores.
- The application was commercially adopted and ran as the primary management system for **two retail shops for over 2 years**, handling inventory, sales, and reporting.

Mobile Application Developer |

Dhaka, Bangladesh | c. 2011–2013

- Developed and shipped 5-10+ mobile applications for the Apple App Store, contributing to a high-volume corporate production pipeline.
- Specialized in iOS development using **Xcode and Objective-C**.
- Expanded skillset to include cross-platform game development using **Unity3D** (C# scripting) for Android and **Flash (ActionScript)**.

Independent iOS Developer | Freelance

Dhaka, Bangladesh | 2010–2011

- Began a career as an independent iOS game developer, successfully developing and launching 5-7 games on the Apple App Store.
- Managed the entire product lifecycle, from concept and coding (Objective-C, Cocos2D) to debugging and App Store submission.

Selected Technical Projects & Portfolio

- **MedTech Information App (Laravel) |**

- Full-stack web application (initially built in **Django**, rebuilt in **Laravel**) to check medicine

information and allow users to manage and track medication-taking habits.

- **BDJobsGuru - Exam Preparation (Laravel)** | bdjobsguru.com
 - Developed and maintained a popular e-learning web application for students and job-seekers preparing for employment and admission exams.
- **Mess Management App (MERN Stack)** | mess.bdjobsguru.com
 - Architected a full-stack **MERN** (MongoDB, Express, React, Node.js) application for mess/mess management.
- **Google Play & App Store Portfolio**
 - **Flutter:** TicTacToe (available on Google Play Store), Ncdms RIC, b2Mapp
 - **Legacy iOS (Objective-C/Cocos2D):** Kids Origami, Birds Adventure, Shadow Bird, Puzzle Match, Jumping Frog, Fun Paint, Clumsy Ball, and others.

List of Publications

1. A. S. M. Latiful Hoque, M. R. Mia, M. S. Abdullah, M. J. Islam, **B. C. D. Nath**, & M. T. Rahman. (2024). "BlockPRLS: Blockchain-Based Patient Record Linkage System for Big Data Analytics." *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, Osaka, Japan, pp. 877-886.
2. R. Mia, A. H. Zahid, **B. C. D. Nath**, & A. S. M. Latiful Hoque. (2021). "A Conceptual Design of Virtual Internship System to Benchmark Software Development Skills in a Blended Learning Environment." *2020 23rd International Conference on Computer & Information Technology (ICCIT)*, 1-6.
3. G. M. Muradul Bashir, A. S. M. Latiful Hoque, & **B. C. D. Nath**. (2016). "E-learning of PHP based on the solutions of real-life problems." *Journal of Computers in Education*, 3, 105-129.

Education

M. Eng. in Information and Communication Technology (ICT) | 2015

Institutes of Information and Communication Technology (IICT), BUET, Dhaka, Bangladesh

M. Sc. in Mathematics (2nd Class) | 2009

National University, Bangladesh

Postgraduate Diploma (PGD) in ICT | 2006

Institutes of Information and Communication Technology (IICT), BUET, Dhaka, Bangladesh

B. Sc. in Mathematics (2nd Class) | 2005

National University, Bangladesh

Personal Information

- **Date of Birth:** 06 Jan 1983
- **Nationality:** Bangladeshi
- **Marital Status:** Married
- **Home district:** Chattogram